

Mechanical Properties & Microstructures of Carbon Steels and Copper Alloys

Materials research using the ASM Handbooks: property data and reference micrographs for seven engineering alloys

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Course	ME 215, NJIT	Type	Individual research project
Sources	ASM Handbooks (ASM International); DoITPoMS Micrograph Library	Materials	AISI 1018, 1040, 1080; copper; bronze; brass; free-machining brass

PROJECT SUMMARY

This research project compiled mechanical and physical property data for three plain-carbon steels and four copper-based materials using the ASM Handbooks, and paired each material with a reference micrograph. The data show two clear trends. In the steels, increasing carbon content from 0.18% (1018) to 0.80% (1080) raises tensile strength by nearly 2× while cutting elongation by more than half, and cold drawing raises strength further at the expense of ductility. In the copper alloys, work hardening increases strength sharply while reducing elongation, and alloying copper with tin or zinc significantly lowers electrical and thermal conductivity.

1. Objective & Method

The goal was to build a side-by-side reference of common engineering alloys that links processing condition, mechanical behavior, and microstructure. Properties were located in the ASM Handbook series through the NJIT Library and organized into tables by material family. For each material, a representative micrograph was located and attributed in APA style.

2. Carbon Steels

Table 1. Mechanical properties of plain-carbon steel bars (ASM International, 2002)

Steel	Condition	Tensile strength MPa (ksi)	Yield strength (MPa)	Elongation (%)	Reduction of area (%)	Hardness (HB)
AISI 1018 (low carbon)	Hot rolled	400 (58)	220	25	50	116
	Cold drawn	440 (64)	370	15	40	126
AISI 1040 (medium carbon)	Hot rolled	520 (76)	290	18	40	149
	Cold drawn	590 (85)	490	12	35	170
AISI 1080 (high carbon)	Hot rolled	770 (112)	420	10	25	229
	Spheroidized, cold drawn	680 (98)	520	10	45	192

3. Copper and Copper Alloys

Table 2. Mechanical properties of copper and copper alloys

Material	Condition	Tensile strength (MPa)	Yield strength (MPa)	Elongation (%)	Melting range (°C)
Copper	Hard	344	310	6	1065–1080
	Hot rolled	234	68	45	
Bronze	Hard	558	517	10	955–1050
Brass	Hard	510	413	8	905–935
	Annealed	337	120	57	
Free-machining brass	—	296	213	16	885–900

Table 3. Physical properties of copper and copper alloys

Material	Electrical conductivity (% IACS)	Thermal conductivity (cal/cm·s·°C)	Coefficient of thermal expansion (μin/in·°F)
Copper	97–100	0.9	9.3
Bronze	15	0.17	9.9
Brass	27	0.28	10.2
Free-machining brass	26	0.27	11.4

4. Observations

Carbon content. Moving from 1018 to 1080 steel nearly doubles tensile strength in the hot-rolled condition (400 to 770 MPa) and doubles hardness, while elongation falls from 25% to 10%. More carbon means more pearlite in the microstructure, which strengthens the steel but limits how far it can deform before fracture.

Processing condition. Cold drawing raises the yield strength of 1018 from 220 to 370 MPa while reducing elongation from 25% to 15%, a direct result of work hardening. The same trade-off appears in copper and brass: hard (cold-worked) tempers are several times stronger in yield than their hot-rolled or annealed counterparts but have a fraction of the ductility. Spheroidizing 1080 before drawing lowers its hardness and improves its reduction of area, which is why that treatment is used to improve machinability.

Alloying and conductivity. Pure copper has the highest electrical and thermal conductivity of the group. Adding tin (bronze) or zinc (brass) scatters electrons and cuts conductivity to roughly 15–27% IACS, but produces much higher strength — which is why pure copper is chosen for electrical applications and its alloys for structural and wear parts.

5. Reference Micrographs



Figure 1. AISI 1018 low-carbon steel



Figure 2. AISI 1040 medium-carbon steel



Figure 3. AISI 1080 high-carbon steel

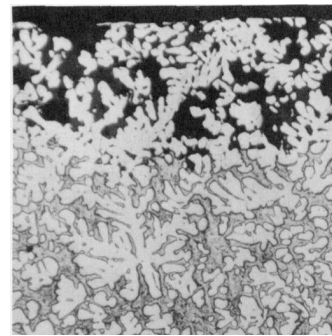


Figure 4. Copper

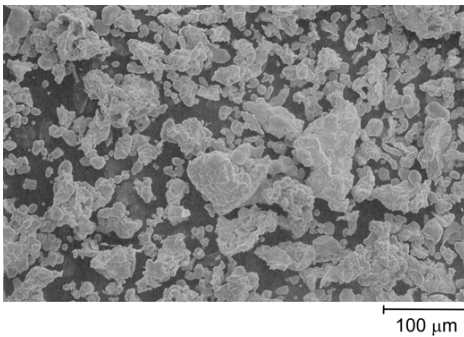


Figure 5. Bronze

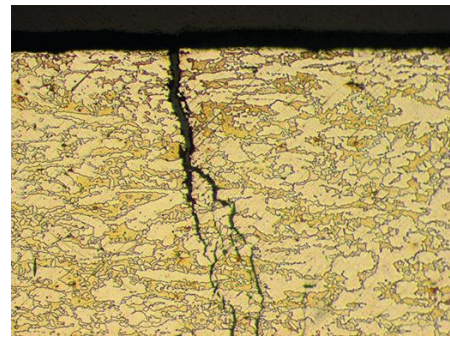


Figure 6. Brass

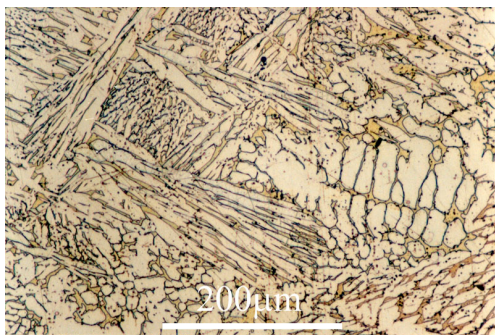


Figure 7. Free-machining brass

Micrographs are reproduced for educational purposes from the ASM Handbook volumes and the DoITPoMS Micrograph Library listed in the references.

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